

Analysis 2
Serie 13
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1.

$$y''(x) + 4y'(x) + 4y(x) = 2e^{-2x}$$

$$\lambda^2 + 4\lambda + 4 = 0 \Rightarrow \lambda = -2$$

$$y_h = (C_1 + xC_2)e^{-2x}$$

$$y_p = C_3x^2e^{-2x}$$

$$y'_p = C_3e^{-2x}(2x - 2x^2)$$

$$y''_p = C_3e^{-2x}(2 - 4x - 4x + 4x^2) = C_3e^{-2x}(2 - 8x + 4x^2)$$

$$\Rightarrow y''_p + 4y'_p + 4y_p = C_3e^{-2x}(2 - 8x + 4x^2 + 8x - 8x^2 + 4x^2) = C_3e^{-2x}2 = e^{-2x}$$

$$\Rightarrow C_3 = 1$$

$$y(0) = (C_1 + xC_2 + x^2)e^{-2x} = C_1 = 0$$

$$y'(0) = C_2 = 0$$

$$\Rightarrow y(x) = x^2e^{-2x}$$

2.

2.a.

$$2y'' - 24y' + 72y = 0$$

$$2\lambda^2 - 24\lambda + 72 = 0 \Rightarrow \lambda = 6$$

$$y(x) = (C_1 + xC_2)e^{6x} \quad C_1 = 1 \quad C_2 = 4$$

2.b.

$$\lambda^2 + \lambda - 6 = 0 \Rightarrow \lambda \in \{2, -3\}$$

$$y(x) = C_1e^{2x} + C_2e^{-3x}$$

$$C_1 + C_2 = 1$$

$$2C_1 - 3C_2 = 7$$

$$\Rightarrow C_1 = 2 \quad C_2 = -2$$

2.c.

$$\lambda^2 - 2\lambda + 2 = 0$$

$$\lambda_{1,2} = \frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i = \sqrt{2}e^{\pm \frac{\pi}{4}i}$$

$$y(x) = C_1e^{x(1+i)} + C_2e^{x(1-i)} = e^x(C_1(i\sin(x) + \cos(x)) + C_2(i\sin(-x) + \cos(-x)))$$

$$= e^x((C_1 + C_2)\cos(x) + i(C_1 - C_2)\sin(x)) = e^x(A\cos(x) + B\sin(x))$$

With $A = C_1 + C_2$ and $B = i(C_1 - C_2)$

$$y(0) = A = 0$$

$$y'(0) = A + B = -1 \Rightarrow B = -1$$

3.

3.a.

$$\begin{aligned}y_h &= (C_1 + xC_2)e^{-2x} \\y_p &= A \cos + B \sin \\y'_p &= -A \sin + B \cos \\y''_p &= -A \cos - B \sin \\-A \cos - B \sin - 4(-A \sin + B \cos) + 4(A \cos + B \sin) &= \sin \\3A - 4B &= 0 \\4A + 3B &= 1 \\\Rightarrow A &= \frac{4}{25} \quad B = \frac{3}{25} \\y(x) &= (C_1 + xC_2)e^{-2x} + \frac{4}{25} \cos(x) + \frac{3}{25} \sin(x)\end{aligned}$$

3.b.

$$\begin{aligned}y_h &= C_1 e^x + C_2 e^{-x} \\y_p &= Ax + B \quad y'_p = A \quad y''_p = 0 \\0 - Ax - B &= x \Rightarrow B = 0 \quad A = -1 \\y(0) &= C_1 + C_2 + 0 = 1 \\y'(0) &= C_1 - C_2 - 1 = 3 \\\Rightarrow C_1 &= \frac{5}{2} \quad C_2 = -\frac{3}{2} \\y(x) &= \frac{5}{2} e^x - \frac{3}{2} e^{-x} - x\end{aligned}$$

3.c.

$$\begin{aligned}H(x) &= \int H'(x) dx \int -\frac{4}{x} - 1 dx = -4 \ln(x) - x \\\frac{d}{dx}(e^{H(x)} y(x)) &= e^{H(x)} (y' + H'(x)y) \\\Rightarrow y(x) &= e^{-H(x)} \int e^{H(x)} x^4 dx \\&= x^4 e^x \int x^{-4} e^{-x} x^4 dx = x^4 e^x (C - e^{-x}) \\y(1) &= e \left(C - \frac{1}{e} \right) = 1 \Rightarrow C = 0 \\y(x) &= -x^4\end{aligned}$$

4.

4.a.

Problems with $x \in \{0, 1\}$. Everywhere else fine.

$$\begin{aligned}
\frac{dx}{dy} &= f(x, y) = \frac{y^2 + y}{x^2 - x} \\
\Rightarrow \int \frac{1}{y^2 + y} dy &= \int \frac{1}{x^2 - x} dx \\
\int \frac{1}{y} - \frac{1}{y+1} dy &= \int \frac{1}{x-1} - \frac{1}{x} dx \\
\ln(y) - \ln(y+1) &= \ln(x-1) - \ln(x) + C \\
\frac{y}{y+1} &= C' \frac{x-1}{x} \\
y \left(1 - C' \frac{x-1}{x} \right) &= C' \frac{x-1}{x} \\
y &= C' \frac{(x-1)}{x - C'(x-1)}
\end{aligned}$$

4.b.

$$\begin{aligned}
\frac{dy}{dx} &= 1 - e^y \\
\int \frac{1}{1 - e^y} dy &= \int 1 dx \\
\int \frac{1}{u(u-1)} du &= x + C' \\
\frac{e^y}{1 - e^y} &= C e^x \\
y(x) &= \ln \left(\frac{C e^x}{1 + C e^x} \right)
\end{aligned}$$

4.c.

$$\begin{aligned}
\int \frac{1}{1-y^2} dy &= \int \frac{1}{x} dx \\
\int \frac{\frac{1}{2}}{y+1} + \frac{\frac{1}{2}}{1-y} dy &= \ln(x) + C' \\
\frac{1}{2} \ln(y+1) - \frac{1}{2} \ln(1-y) &= \ln(x) + C' \\
\frac{1+y}{1-y} &= C x^2 \\
y(x) &= \frac{C x^2 - 1}{C x^2 + 1}
\end{aligned}$$

5.

$$y' = \left(\frac{y}{x} \right)^2 + \frac{y}{x} + 1$$

Let $v = \frac{y}{x}$

$$vx = y \Rightarrow y' = v'x + x'v = v'x + v$$

then

$$y' = \left(\frac{y}{x}\right)^2 + \frac{y}{x} + 1$$

$$\Leftrightarrow v'x + v = v^2 + v + 1$$

$$\int \frac{1}{v^2 + 1} dv = \int \frac{1}{x} dx$$

$$\int \frac{\cos^2(v)}{\tan^2(v) + 1} du = \int 1 du = \ln(x) + C$$

$$u = \tan(v) = \ln(x) + C$$

$$y(x) = vx = x \tan^{-1}(\ln(x) + C)$$

5.a.

(a) $y(x) = Ce^{2x}$

true

(b) $y(x) = (C \pm \sqrt{2x})^2$

false

(c) $y(x) = \left(\frac{x^2}{4} + C\right)^2$

false

(d) $y(x) = Ce^x - 1$

false

(e) $y(x) = Ce^{\frac{x^2}{3}}$

true

5.b.

(a) false

(b) true

(c) false

(d) false

5.c.

(a) true

(b) false

(c) true

6.

(a) false

(b) true

(c) false

(d) false